LESSON 4 • TERM 1 • LEVEL 1

100 Square Explorer

Whole numbers: patterns in rows and columns

TODAY WE WILL

Explore the 100 square and find the patterns hiding in its rows and columns.



Get ready

0–4 MIN

- Your 100 square (the kit page)
- About 12 counters, all one kind
- Two or three colours
- Workbook, pencil and paper

Counters stay on the table, never near the mouth or ears.

4–10 MIN

Warm-up: count in tens

Show both hands for every ten

10	20	30	40	50	60	70	80	90	100

Now count back from 100. Whisper, and shout on 50!

Count along with me!



4–10 MIN

Fives and twos

One hand is five fingers. Count in fives:

5 10 15 20 25 30

Socks come in pairs. A pair is two. Count in twos:

2 4 6 8 10 12

4–10 MIN

What comes next?

10 20 30 ___	5 10 15 ___	2 4 6 ___	28 29 ___	38 39 ___

Say the next number. Show it on your fingers if you like.

Sneak-peek puzzles

10–14 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Count in 5s. What are the next two numbers?

5	10	15	20	___	___

PUZZLE 2: think

This is part of the 100 square. A counter is on 36.

Ravi moves it down one square.

Which number is it on now? Write how you know.

21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

Explore: read the square

14–28 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1. Put a finger on 1. Slide it along to 10.
2. Where does 11 sit? Slide your finger down.
3. Find 100. Which corner is it in?

Explore: cover the tens

14–28 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1. Count in tens: 10, 20, 30, ...
2. Put a counter on each number you say.
3. What do you see?

SAY IT

The counters make a line that goes ___ (across or down).

Explore: fives and twos

14–28 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

1. Clear your counters. Count in fives to 50.
2. What do you see?
3. Clear again. Count in twos to 20.
4. Which numbers have no counter?

Counting in 2s: will 37 get a counter? Will 60?

Explore: move the counter

14–28 MIN

11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

1. Counter on 34. Move it right 1.
2. Now down 1. Down 1 again.
3. Left 1. Then up 1.

WHAT DO YOU NOTICE?

Which way adds 1?

Which way adds 10?

28–38 MIN

Across: 1 more, 1 less

21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50





Right: 1 more. Left: 1 less.

40 is the last number in its row. The next number, 41, starts the next row.

Down: 10 more, up: 10 less

28–38 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100



Down: 10 more. Up: 10 less.

A row has ten squares. So one step down is ten more.

Your turn: write the number

28–38 MIN

Write on your paper. Timer: 1 minute

1

10 more than 36

2

10 less than 58

3

1 more than 40

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

28–38 MIN

Check: write the number

1

10 more than 36

46

2

10 less than 58

48

3

1 more than 40

41

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Stripes and last digits

28–38 MIN

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

10s: one column

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

5s: two columns

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

2s: stripes

THE LAST DIGIT

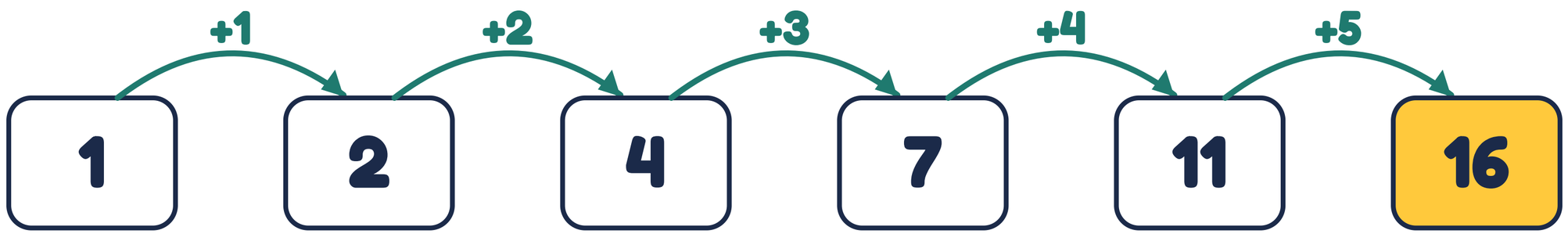
Counting in 10s: every number ends in 0.

Counting in 5s: every number ends in 0 or 5.

Counting in 2s: every number ends in 0, 2, 4, 6 or 8.

28–38 MIN

Gaps can grow



The gap grows by 1 each time. So the next gap is 5.

Quick-fire: 10 more

28–38 MIN

23 47 **65** 90

Say the number that is 10 more. Think: one row down.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

38–42 MIN

Jump, clap, tap

A big jump for every ten. A clap for every five. A tiny tap for every two.

10 5 2

Sip some water.

Workbook page 1: Start here

42–70 MIN

Questions 1 and 2. Timer: 8 minutes

S1 Colour 10, 20, 30 and so on to 100 as you count in 10s. Fill in the blanks.

S2 Count in 5s. Colour 5, 10, 15 and keep going to 100. Fill in the blanks.

Check page 1

42–70 MIN

S1	They end in 0. They make a line going down.
S2	They end in 5 or 0. They make 2 lines going down.

Workbook page 2: Start here

42–70 MIN

Questions 3 to 6. Timer: 9 minutes

S3 What are the next two numbers in each of these patterns?

S4 Count in 2s. Colour 2, 4, 6 and so on to 30. Look at the last digits.

S5 Fill in the missing number in each empty square.

S6 Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it.

Check page 2

42–70 MIN

S3	10, 12 | 30, 35 | 60, 70 | 41, 51
S4	2, 4, 6, 8, 0 (any order)
S5	a 34, b 43, c 55
S6	26 is wrong. She should have said 25.

42–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Our idea in a new setting
Go further (page 4)	Bigger puzzles
Investigation (page 5)	Counting in 3s
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

TALK ABOUT IT

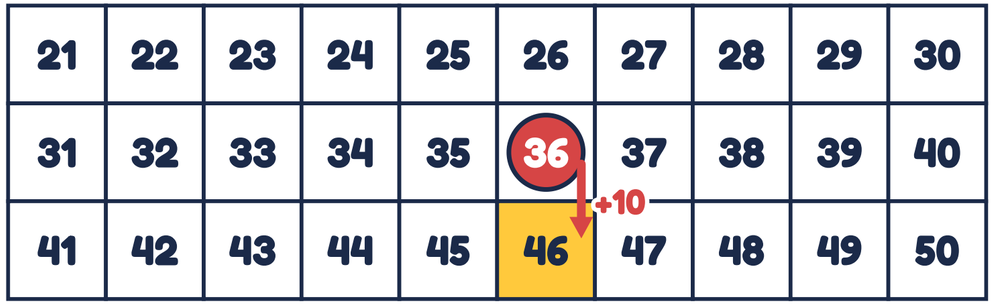
Does counting in 5s make one column, like counting in 10s?

Use the 100 square to show why.

I think ... because ...

Back to the sneak-peek

70–76 MIN



The counter is on 46. Down one row is 10 more.

Exit ticket

76–82 MIN

On your own. No looking back. Timer: 3 minutes

E1 **Count in 5s. What are the next two numbers?**

35, 40, 45, _____, _____

E2 **A counter is on 26 on your 100 square. First it moves down 1 row. Then it moves right 1 square.**

(a) After going down it is on _____.

(b) After going right it is on _____.

E3 **Mei counts in 2s: 2, 4, 6, 8 and so on. Will she say 25?**

Circle Yes or No.

Check the exit ticket

76–82 MIN

E1	50, 55
E2	36, then 37
E3	No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8.

Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ finding my place on the square ☐ counting in 2s, 5s or 10s ☐ moving down a row ☐ the pattern where the gap grows (Think harder) ☐ something else
What helped me	☐ the 100 square or counters ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck
Next time I will try	☐ drawing it ☐ using counters ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

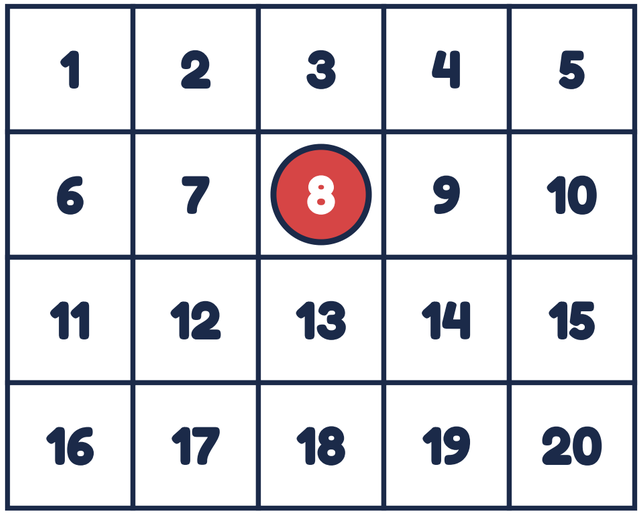
Adding by making tens

Have ready: 20 small things, kit page cut out.



Go further: what if?

OPTIONAL



Here is a square with only 5 numbers in each row.

A counter is on 8. It moves down 1 row. Which number is it on?

What does going down add now?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Add 10 to a number. Does only the tens digit change?
Make a puzzle	Hide a number on the square and write clues for your grown-up
Teach it back	Show why counting in 10s makes a column

Optional. Everyone may join.